

Differential Equations (2015–2021)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

44

TOTAL MARKS

75 M

TEST DURATION

132 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	23	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	6	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	15	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q48 • ID: JAM_2021_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let $y : \left(\frac{9}{10}, 3\right) \rightarrow \mathbb{R}$ be a differentiable function satisfying

$$(x - 2y)\frac{dy}{dx} + (2x + y) = 0, \quad x \in \left(\frac{9}{10}, 3\right), \quad \text{and } y(1) = 1.$$

Then $y(2)$ equals ____.

Question 2 (JAM 2021 • Q19 • ID: JAM_2021_Q19)

MCQ

+2 M

Neg: -0.67

Q. 19 Consider the family of curves $x^2 - y^2 = ky$ with parameter $k \in \mathbb{R}$. The equation of the orthogonal trajectory to this family passing through $(1, 1)$ is given by

(A) $x^3 + 3xy^2 = 4$.

(B) $x^2 + 2xy = 3$.

(C) $y^2 + 2x^2y = 3$.

(D) $x^3 + 2xy^2 = 3$.

Question 3 (JAM 2021 • Q12 • ID: JAM_2021_Q12)

MCQ

+2 M

Neg: -0.67

Q. 12 Let y be the solution of

$$(1+x)y''(x) + y'(x) - \frac{1}{1+x}y(x) = 0, \quad x \in (-1, \infty),$$

$$y(0) = 1, \quad y'(0) = 0.$$

Then

(A) y is bounded on $(0, \infty)$.(B) y is bounded on $(-1, 0]$.(C) $y(x) \geq 2$ on $(-1, \infty)$.(D) y attains its minimum at $x = 0$.

Question 4 (JAM 2020 • Q57 • ID: JAM_2020_Q57)

NAT

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 57 If $x^2 + xy^2 = c$, where $c \in \mathbb{R}$, is the general solution of the exact differential equation

$$M(x, y) dx + 2xy dy = 0,$$

then $M(1, 1)$ is _____.

$$\begin{bmatrix} 9 & 2 & 7 & 1 \end{bmatrix}$$

Question 5 (JAM 2020 • Q56 • ID: JAM_2020_Q56)

NAT

+2 M

Neg: Nil (0)

Q. 56 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f'(x) = f(x)$ for all x . Suppose that $f(\alpha x)$ and $f(\beta x)$ are two non-zero solutions of the differential equation

$$4 \frac{d^2 y}{dx^2} - p \frac{dy}{dx} + 3y = 0$$

satisfying

$$f(\alpha x)f(\beta x) = f(2x) \text{ and } f(\alpha x)f(-\beta x) = f(x).$$

Then, the value of p is _____.

MA

16 / 17

Question 6 (JAM 2020 • Q48 • ID: JAM_2020_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let $f(x, y) = 0$ be a solution of the homogeneous differential equation

$$(2x + 5y)dx - (x + 3y)dy = 0.$$

If $f(x + \alpha, y - 3) = 0$ is a solution of the differential equation

$$(2x + 5y - 1)dx + (2 - x - 3y)dy = 0,$$

then the value of α is _____.

Question 7 (JAM 2020 • Q46 • ID: JAM_2020_Q46)

NAT

+1 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 46 Consider the differential equation

$$\frac{dy}{dx} + 10y = f(x), \quad x > 0,$$

where $f(x)$ is a continuous function such that $\lim_{x \rightarrow \infty} f(x) = 1$. Then the value of

$$\lim_{x \rightarrow \infty} y(x)$$

is _____.

Question 8 (JAM 2020 • Q35 • ID: JAM_2020_Q35)

MSQ

+2 M

Neg: Nil (0)

Q. 35 Let $L[y] = x^2 \frac{d^2 y}{dx^2} + px \frac{dy}{dx} + qy$, where p, q are real constants. Let $y_1(x)$ and $y_2(x)$ be two solutions of $L[y] = 0, x > 0$, that satisfy $y_1(x_0) = 1, y_1'(x_0) = 0, y_2(x_0) = 0$ and $y_2'(x_0) = 1$ for some $x_0 > 0$. Then,

- (A) $y_1(x)$ is not a constant multiple of $y_2(x)$
- (B) $y_1(x)$ is a constant multiple of $y_2(x)$
- (C) $1, \ln x$ are solutions of $L[y] = 0$ when $p = 1, q = 0$
- (D) $x, \ln x$ are solutions of $L[y] = 0$ when $p + q \neq 0$

Question 9 (JAM 2020 • Q22 • ID: JAM_2020_Q22)

MCQ

+2 M

Neg: -0.67

Q. 22 A solution of the differential equation $2x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} - y = 0, x > 0$ that passes through the point $(1, 1)$ is

- (A) $y = \frac{1}{x}$
- (B) $y = \frac{1}{x^2}$
- (C) $y = \frac{1}{\sqrt{x}}$
- (D) $y = \frac{1}{x^{3/2}}$

Question 10 (JAM 2020 • Q18 • ID: JAM_2020_Q18)

MCQ

+2 M

Neg: -0.67

Q. 18 The area bounded by the curves $x^2 + y^2 = 2x$ and $x^2 + y^2 = 4x$, and the straight lines $y = x$ and $y = 0$ is

- (A) $3\left(\frac{\pi}{2} + \frac{1}{4}\right)$ (B) $3\left(\frac{\pi}{4} + \frac{1}{2}\right)$ (C) $2\left(\frac{\pi}{4} + \frac{1}{3}\right)$ (D) $2\left(\frac{\pi}{3} + \frac{1}{4}\right)$

Question 11 (JAM 2020 • Q12 • ID: JAM_2020_Q12)

MCQ

+2 M

Neg: -0.67

Q. 12 Define $s_1 = \alpha > 0$ and $s_{n+1} = \sqrt{\frac{1+s_n^2}{1+\alpha}}$, $n \geq 1$. Which of the following is true?

- (A) If $s_n^2 < \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\sqrt{\alpha}}$
 (B) If $s_n^2 < \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\alpha}$
 (C) If $s_n^2 > \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically increasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\sqrt{\alpha}}$
 (D) If $s_n^2 > \frac{1}{\alpha}$, then $\{s_n\}$ is monotonically decreasing and $\lim_{n \rightarrow \infty} s_n = \frac{1}{\alpha}$

MA

5 / 17

Question 12 (JAM 2020 • Q7 • ID: JAM_2020_Q7)

MCQ

+1 M

Neg: -0.33

Q. 7 If $u = x^3$ and $v = y^2$ transform the differential equation $3x^5 dx - y(y^2 - x^3)dy = 0$ to $\frac{dv}{du} = \frac{\alpha u}{2(u-v)}$, then α is

- (A) 4 (B) 2 (C) -2 (D) -4

Question 13 (JAM 2020 • Q3 • ID: JAM_2020_Q3)

MCQ

+1 M

Neg: -0.33

Q. 3 Which of the following is FALSE?

(A) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$

(B) $\lim_{x \rightarrow 0^+} \frac{1}{xe^{1/x}} = 0$

(C) $\lim_{x \rightarrow 0^+} \frac{\sin x}{1 + 2x} = 0$

(D) $\lim_{x \rightarrow 0^+} \frac{\cos x}{1 + 2x} = 0$

Question 14 (JAM 2019 • Q60 • ID: JAM_2019_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If $y(x)$ is the solution of the initial value problem

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0, \quad y(0) = 2, \quad \frac{dy}{dx}(0) = 0,$$

then $y(\ln 2)$ is (round off to 2 decimal places) equal to _____

END OF THE QUESTION PAPER

Question 15 (JAM 2019 • Q59 • ID: JAM_2019_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 Let $y(x) = xv(x)$ be a solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 3y = 0.$$

If $v(0) = 0$ and $v(1) = 1$, then $v(-2)$ is equal to _____

MA

12/13

Question 16 (JAM 2019 • Q29 • ID: JAM_2019_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 For $\beta \in \mathbb{R}$, define

$$f(x, y) = \begin{cases} \frac{x^2 |x|^\beta y}{x^4 + y^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then, at $(0, 0)$, the function f is

- (A) continuous for $\beta = 0$
- (B) continuous for $\beta > 0$
- (C) not differentiable for any β
- (D) continuous for $\beta < 0$

Question 17 (JAM 2019 • Q19 • ID: JAM_2019_Q19)

MCQ

+2 M

Neg: -0.67

Q.19 Which one of the following series is divergent?

(A) $\sum_{n=1}^{\infty} \frac{1}{n} \sin^2 \frac{1}{n}$

(C) $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{1}{n}$

(B) $\sum_{n=1}^{\infty} \frac{1}{n} \log n$

(D) $\sum_{n=1}^{\infty} \frac{1}{n} \tan \frac{1}{n}$

MA

5/13

Question 18 (JAM 2019 • Q12 • ID: JAM_2019_Q12)

MCQ

+2 M

Neg: -0.67

Q.12 Let P be a 4×4 matrix with entries from the set of rational numbers. If $\sqrt{2} + i$, with $i = \sqrt{-1}$, is a root of the characteristic polynomial of P and I is the 4×4 identity matrix, then

- (A) $P^4 = 4P^2 + 9I$ (B) $P^4 = 4P^2 - 9I$ (C) $P^4 = 2P^2 - 9I$ (D) $P^4 = 2P^2 + 9I$

MA

4/13

Question 19 (JAM 2019 • Q8 • ID: JAM_2019_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 The equation of the tangent plane to the surface $x^2z + \sqrt{8 - x^2 - y^4} = 6$ at the point $(2, 0, 1)$ is

- (A) $2x + z = 5$ (B) $3x + 4z = 10$
(C) $3x - z = 10$ (D) $7x - 4z = 10$

Question 20 (JAM 2018 • Q60 • ID: JAM_2018_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If $y(x) = v(x) \sec x$ is the solution of $y'' - (2 \tan x) y' + 5y = 0$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, satisfying $y(0) = 0$ and $y'(0) = \sqrt{6}$, then $v\left(\frac{\pi}{6\sqrt{6}}\right)$ is _____ (correct up to two decimal places).

END OF THE QUESTION PAPER

MA

12/12

Question 21 (JAM 2018 • Q59 • ID: JAM_2018_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 If $x(t)$ is the solution to the differential equation $\frac{dx}{dt} = x^2t^3 + xt$, for $t > 0$, satisfying $x(0) = 1$, then the value of $x(\sqrt{2})$ is _____ (correct up to two decimal places).

Question 22 (JAM 2018 • Q46 • ID: JAM_2018_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46 Let $f: [0, \infty) \rightarrow [0, \infty)$ be continuous on $[0, \infty)$ and differentiable on $(0, \infty)$. If

$$f(x) = \int_0^x \sqrt{f(t)} dt, \text{ then } f(6) = \underline{\hspace{2cm}}$$

MA

10/12

Question 23 (JAM 2018 • Q32 • ID: JAM_2018_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32 The solution(s) of the differential equation $\frac{dy}{dx} = (\sin 2x) y^{1/3}$ satisfying $y(0) = 0$ is (are)

(A) $y(x) = 0$

(B) $y(x) = -\sqrt{\frac{8}{27}} \sin^3 x$

(C) $y(x) = \sqrt{\frac{8}{27}} \sin^3 x$

(D) $y(x) = \sqrt{\frac{8}{27}} \cos^3 x$

Question 24 (JAM 2018 • Q28 • ID: JAM_2018_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 Consider the group $\mathbb{Z}^2 = \{(a, b) \mid a, b \in \mathbb{Z}\}$ under component-wise addition. Then which of the following is a subgroup of $\mathbb{Z}^2$?

(A) $\{(a, b) \in \mathbb{Z}^2 \mid ab = 0\}$

(B) $\{(a, b) \in \mathbb{Z}^2 \mid 3a + 2b = 15\}$

(C) $\{(a, b) \in \mathbb{Z}^2 \mid 7 \text{ divides } ab\}$

(D) $\{(a, b) \in \mathbb{Z}^2 \mid 2 \text{ divides } a \text{ and } 3 \text{ divides } b\}$

Question 25 (JAM 2018 • Q18 • ID: JAM_2018_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 Let $a, b \in \mathbb{R}$ and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. If $z = e^u f(v)$, where $u = ax + by$ and $v = ax - by$, then which one of the following is TRUE?

(A) $b^2 z_{xx} - a^2 z_{yy} = 4a^2 b^2 e^u f'(v)$

(B) $b^2 z_{xx} - a^2 z_{yy} = -4e^u f'(v)$

(C) $b z_x + a z_y = ab z$

(D) $b z_x + a z_y = -ab z$

Question 26 (JAM 2018 • Q8 • ID: JAM_2018_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 In $\mathbb{R}^2$, the family of trajectories orthogonal to the family of asteroids $x^{2/3} + y^{2/3} = a^{2/3}$ is given by

(A) $x^{4/3} + y^{4/3} = c^{4/3}$

(B) $x^{4/3} - y^{4/3} = c^{4/3}$

(C) $x^{5/3} - y^{5/3} = c^{5/3}$

(D) $x^{2/3} - y^{2/3} = c^{2/3}$

Question 27 (JAM 2017 • Q60 • ID: JAM_2017_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 If

$$y(x) = \int_{\sqrt{x}}^x \frac{e^t}{t} dt, \quad x > 0$$

Question 28 (JAM 2017 • Q51 • ID: JAM_2017_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51 Let $y(x)$, $x > 0$ be the solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0$$

satisfying the conditions $y(1) = 1$ and $y'(1) = 0$. Then the value of $e^2 y(e)$ is _____

Question 29 (JAM 2017 • Q41 • ID: JAM_2017_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41 If the orthogonal trajectories of the family of ellipses $x^2 + 2y^2 = c_1$, $c_1 > 0$, are given by $y = c_2 x^\alpha$, $c_2 \in \mathbb{R}$, then $\alpha =$ _____

Question 30 (JAM 2017 • Q40 • ID: JAM_2017_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Let $k, \ell \in \mathbb{R}$ be such that every solution of

$$\frac{d^2 y}{dx^2} + 2k \frac{dy}{dx} + \ell y = 0$$

satisfies $\lim_{x \rightarrow \infty} y(x) = 0$. Then

- (A) $3k^2 + \ell < 0$ and $k > 0$
- (B) $k^2 + \ell > 0$ and $k < 0$
- (C) $k^2 - \ell \leq 0$ and $k > 0$
- (D) $k^2 - \ell > 0, k > 0$ and $\ell > 0$

Question 31 (JAM 2017 • Q39 • ID: JAM_2017_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = (y - 1)(y - 3)$$

satisfying the condition $y(0) = 2$. Then which of the following is/are TRUE?

- (A) The function $y(x)$ is not bounded above
- (B) The function $y(x)$ is bounded
- (C) $\lim_{x \rightarrow +\infty} y(x) = 1$
- (D) $\lim_{x \rightarrow -\infty} y(x) = 3$

Question 32 (JAM 2017 • Q30 • ID: JAM_2017_Q30)

MCQ

+2 M

Neg: -0.67

Q.30 Let $y(x)$ be the solution of the differential equation

$$(xy + y + e^{-x})dx + (x + e^{-x})dy = 0$$

satisfying $y(0) = 1$. Then $y(-1)$ is equal to

(A) $\frac{e}{e-1}$

(B) $\frac{2e}{e-1}$

(C) $\frac{e}{1-e}$

(D) 0

Question 33 (JAM 2017 • Q29 • ID: JAM_2017_Q29)

MCQ

+2 M

Neg: -0.67

Q.29 A particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^{2x} \sin x$$

is

(A) $\frac{e^{2x}}{10}(3 \cos x - 2 \sin x)$

(B) $-\frac{e^{2x}}{10}(3 \cos x - 2 \sin x)$

(C) $-\frac{e^{2x}}{5}(2 \cos x + \sin x)$

(D) $\frac{e^{2x}}{5}(2 \cos x - \sin x)$

Question 34 (JAM 2016 • Q56 • ID: JAM_2016_Q56)

NAT

+2 M

Neg: Nil (0)

Q.56 Let $y(x)$ be the solution of the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0, \quad y(0) = 1, \quad \left.\frac{dy}{dx}\right|_{x=0} = -1.$$

Then $y(x)$ attains its maximum value at $x =$ _____

Question 35 (JAM 2016 • Q47 • ID: JAM_2016_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47 Let $y_1(x)$, $y_2(x)$ and $y_3(x)$ be linearly independent solutions of the differential equation

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0.$$

If the Wronskian $W(y_1, y_2, y_3)$ is of the form ke^{bx} for some constant k , then the value of b is _____

Question 36 (JAM 2016 • Q35 • ID: JAM_2016_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35 Consider the differential equation

$$\sin 2x \frac{dy}{dx} = 2y + 2 \cos x, \quad y\left(\frac{\pi}{4}\right) = 1 - \sqrt{2}.$$

Then which of the following statement(s) is(are) TRUE?

- (A) The solution is unbounded when $x \rightarrow 0$
- (B) The solution is unbounded when $x \rightarrow \frac{\pi}{2}$
- (C) The solution is bounded when $x \rightarrow 0$
- (D) The solution is bounded when $x \rightarrow \frac{\pi}{2}$

Question 37 (JAM 2016 • Q25 • ID: JAM_2016_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 Let $f(x) = \frac{1}{1+|x|} + \frac{1}{1+|x-1|}$ for all $x \in [-1, 1]$. Then which one of the following is TRUE?

- (A) Maximum value of $f(x)$ is $\frac{3}{2}$
- (B) Minimum value of $f(x)$ is $\frac{1}{3}$
- (C) Maximum of $f(x)$ occurs at $x = \frac{1}{2}$
- (D) Minimum of $f(x)$ occurs at $x = 1$

Question 38 (JAM 2016 • Q15 • ID: JAM_2016_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 If the triple integral over the region bounded by the planes

$$2x + y + z = 4, \quad x = 0, \quad y = 0, \quad z = 0$$

is given by

$$\int_0^2 \int_0^{\lambda(x)} \int_0^{\mu(x,y)} dz \, dy \, dx,$$

then the function $\lambda(x) - \mu(x, y)$ is

- (A) $x + y$ (B) $x - y$ (C) x (D) y

Question 39 (JAM 2016 • Q8 • ID: JAM_2016_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 The nonzero value of n for which the differential equation

$$(3xy^2 + n^2x^2y)dx + (nx^3 + 3x^2y)dy = 0, \quad x \neq 0,$$

becomes exact is

- (A) -3 (B) -2 (C) 2 (D) 3

Question 40 (JAM 2015 • Q36 • ID: JAM_2015_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.6

The initial value problem

$$y' = \sqrt{y}, \quad y(0) = \alpha, \quad \alpha \geq 0$$

has

- (A) at least two solutions if $\alpha = 0$
 (B) no solution if $\alpha > 0$
 (C) at least one solution if $\alpha > 0$
 (D) a unique solution if $\alpha = 0$

Question 41 (JAM 2015 • Q24 • ID: JAM_2015_Q24)

MCQ

+2 M

Neg: -0.67

Q.24

If $y(t)$ is a solution of the differential equation $y'' + 4y = 2e^t$, then

$$\lim_{t \rightarrow \infty} e^{-t} y(t)$$

is equal to

(A) $\frac{2}{3}$

(B) $\frac{2}{5}$

(C) $\frac{2}{7}$

(D) $\frac{2}{9}$

MA

9/16

Question 42 (JAM 2015 • Q15 • ID: JAM_2015_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 Suppose that the dependent variables z and w are functions of the independent variables x and y , defined by the equations $f(x, y, z, w) = 0$ and $g(x, y, z, w) = 0$, where $f_z g_w - f_w g_z = 1$. Which one of the following is correct?

(A) $z_x = f_w g_x - f_x g_w$

(B) $z_x = f_x g_w - f_w g_x$

(C) $z_x = f_z g_x - f_x g_z$

(D) $z_x = f_z g_w - f_w g_x$

Question 43 (JAM 2015 • Q8 • ID: JAM_2015_Q8)

MCQ

+1 M

Neg: -0.33

Q.8 If an integral curve of the differential equation $(y - x) \frac{dy}{dx} = 1$ passes through $(0, 0)$ and $(\alpha, 1)$, then α is equal to

(A) $2 - e^{-1}$

(B) $1 - e^{-1}$

(C) e^{-1}

(D) $1 + e$

Question 44 (JAM 2015 • Q3 • ID: JAM_2015_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let a, b, c, d be distinct non-zero real numbers with $a + b = c + d$. Then an eigenvalue of the

matrix $\begin{bmatrix} a & b & 1 \\ c & d & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is

(A) $a + c$

(B) $a + b$

(C) $a - b$

(D) $b - d$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Differential Equations (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+1	3
2	MCQ	+2	A
3	MCQ	+2	D
4	NAT	+2	3 to 3
5	NAT	+2	8 to 8
6	NAT	+1	7 to 7
7	NAT	+1	0.1 to 0.1
8	MSQ	+2	A;C
9	MCQ	+2	A
10	MCQ	+2	B
11	MCQ	+2	A
12	MCQ	+1	D
13	MCQ	+1	D
14	NAT	+2	1.12 to 1.25
15	NAT	+2	4 to 4

Q. No.	Type	Marks	Official Key
16	MCQ	+2	B
17	MCQ	+2	B
18	MCQ	+2	C
19	MCQ	+1	B
20	NAT	+2	0.5 to 0.5
21	NAT	+2	−2.80 to −2.70
22	NAT	+1	9 to 9
23	MSQ	+2	A;B;C
24	MCQ	+2	D
25	MCQ	+2	A
26	MCQ	+1	B
27	NAT	+2	1.34 to 1.36
28	NAT	+2	2.9 to 3.1
29	NAT	+1	1.9 to 2.1
30	MSQ	+2	C, D

Q. No.	Type	Marks	Official Key
31	MSQ	+2	B, C, D
32	MCQ	+2	B
33	MCQ	+2	C
34	NAT	+2	−0.3 to −0.25
35	NAT	+1	6.0 to 6.0
36	MSQ	+2	C;D
37	MCQ	+2	A
38	MCQ	+2	D
39	MCQ	+1	D
40	MSQ	+2	A;C
41	MCQ	+2	B
42	MCQ	+2	A
43	MCQ	+1	C
44	MCQ	+1	B

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.